PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION

PROJECT BUTTON HR-V0 ELECTRICAL V3 SRA1-SUPPLY-WATCHDOG CANDIDATE

P1.20 layout preserved; ordinary watchdog contacts removed from PNOZ inputs and series-gate only SRA1 A1; zero safety credit.

01 External listed sources and DC boundaries

File: 01_external_sources.kicad_sch

02 Dual-channel E-stop and RESET eligibility

File: 02_estop_eligibility.kicad_sch

03 Distinct ARM and SRA1 diagnostic supply gate

File: 03_arm_watchdog_eligibility.kicad_sch

04 Contactor coils, mirror contacts and EDM

File: 04_contactor_edm.kicad_sch

05 Redundant actuator-power interruption

File: 05_actuator_interruption.kicad_sch

06 Protected actuator branches and current-limiter carriers

File: 06_branches_and_limiters.kicad_sch

07 Independent watchdog power, controller and drivers

File: 07_watchdog_control.kicad_sch

08 Calculated dual-channel 24 V watchdog feedback

File: 08_watchdog_feedback_interface.kicad_sch

09 Compute and control terminals

File: 09_compute_and_control_terminals.kicad_sch

10 U2D2, DXL star, actuator ports and bonding boundary

File: 10_actuator_interfaces.kicad_sch

11 Watchdog PCB external connectors

File: 11_watchdog_pcb_connectors.kicad_sch

12 Watchdog PCB test access

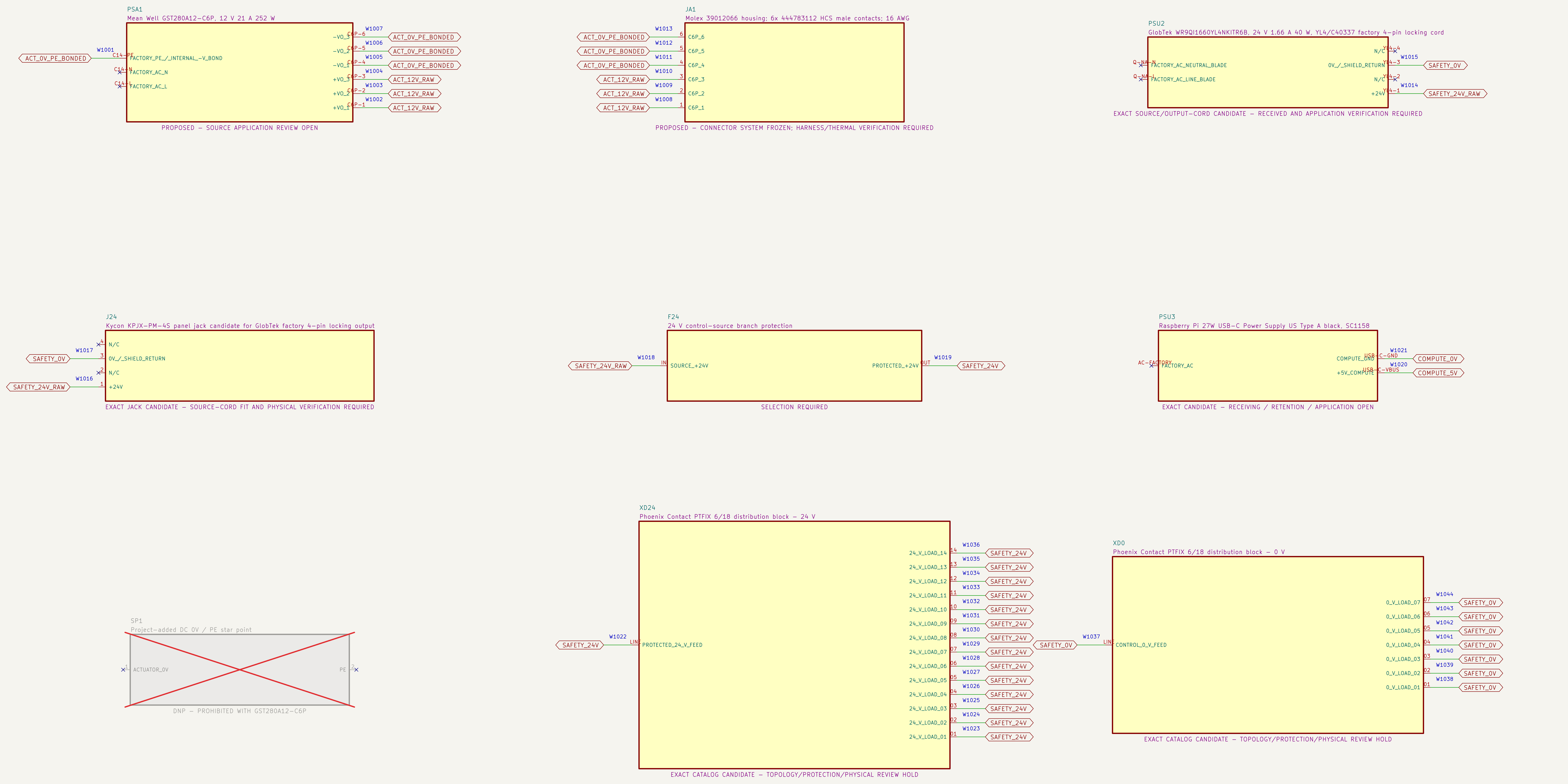
File: 12_watchdog_pcb_test_access.kicad_sch

UNACCEPTED SRA1-SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE
Project Button
Sheet: /
File: project-button-v3-p1.21-sra1-supply-watchdog-candidate.kicad_sch
Title: PB HR-V0 ELEC P1.21 INDEX

Size: A3	Date: 2026-08-11	**Rev: P1.21**
KiCad E.D.A. 10.0.5		Id: 1/13

01 External listed sources and DC boundaries

Factory–sealed adapters eliminate project–built mains wiring.



NOTE 1: All AC conductors shown are factory boundaries, not project–built wiring.

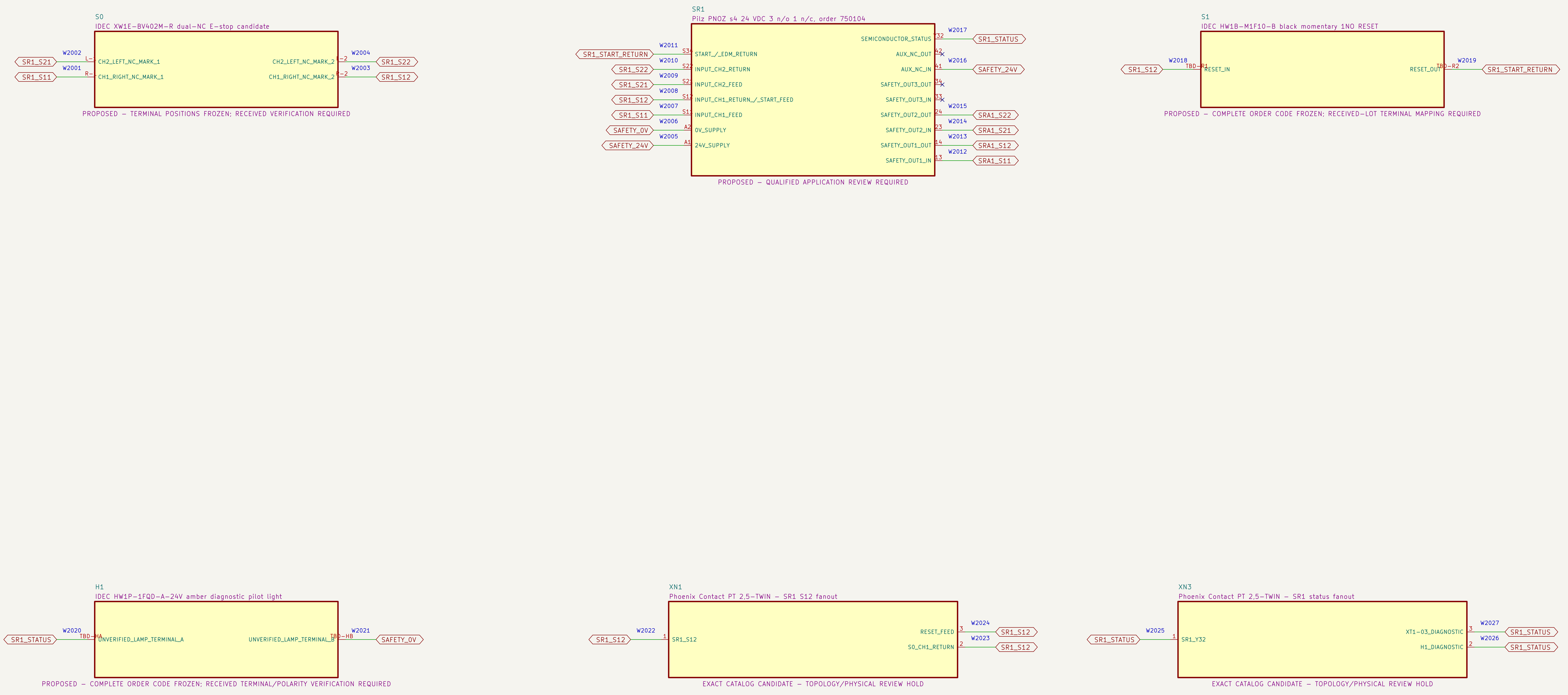
NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

UNACCEPTED SRA1–SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE
Project Button
Sheet: /01 External listed sources and DC boundaries/
File: 01_external_sources.kicad_sch
Title: PB HR–V0 ELEC P1.21 – 01
Size: A2 | Date: 2026–08–11 | **Rev: P1.21**
KiCad E.D.A. 10.0.5 | Id: 2/13

PRELIMINARY — NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION

02 Dual–channel E–stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.



NOTE 1: S0 directly completes both SR1 input returns; ordinary KWD terminals are absent from both loops.

NOTE 2: Heartbeat loss removes only SRA1 A1 through two series ordinary contacts; SR1 remains powered and direct. Repowered SRA1 still requires monitored ARM.

UNACCEPTED SRA1–SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE

Project Button

Sheet: /02 Dual–channel E–stop and RESET eligibility/
File: 02_estop_eligibility.kicad_sch

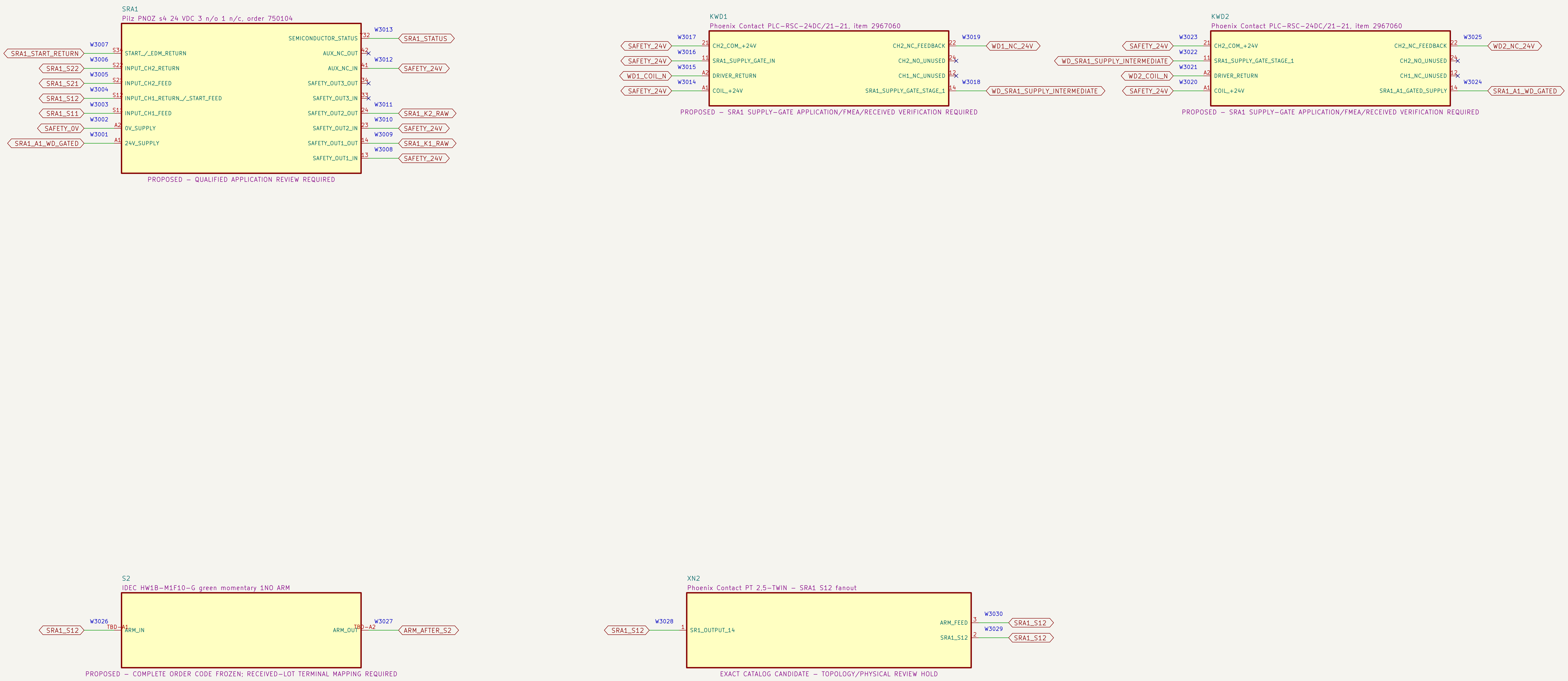
Title: PB HR–V0 ELEC P1.21 — 02

Size: A2	Date: 2026–08–11	**Rev: P1.21**
KiCad E.D.A. 10.0.5		Id: 3/13

PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION

03 Distinct ARM and SRA1 diagnostic supply gate

Two series ordinary KWD contacts gate only SRA1 A1; SR1 and both E–stop input loops remain independent.



NOTE 1: After watchdog dropout: heartbeat healthy –> both KWD contacts restore SRA1 supply –> distinct monitored ARM press/release –> K1/K2 may become eligible. Heartbeat restoration alone must remain OFF.

NOTE 2: KWD1:11–14 and KWD2:11–14 series–gate only SRA1:A1. SR1 supply, S0 input returns and SR1–to–SRA1 input paths remain direct; DF–01 receives zero safety credit.

UNACCEPTED SRA1–SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE
Project Button
Sheet: /03 Distinct ARM and SRA1 diagnostic supply gate/
File: 03_arm_watchdog_eligibility.kicad_sch
Title: PB HR–V0 ELEC P1.21 – 03
Size: A2 | Date: 2026–08–11 | **Rev: P1.21**
KiCad E.D.A. 10.0.5 | Id: 4/13

04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.

NOTE 1: SRA1 outputs are separately protected before K1 and K2 coils. FSR1/FSR2 use proposed Phoenix PT 4−HESI item 3211861 holders with D−ST 4 item 3030420 end cover; both fuse links, grouping and coordination remain SELECTION REQUIRED.

NOTE 2: Loaded DC interruption, suppression behavior, mirror−contact use and coordination require qualified review.

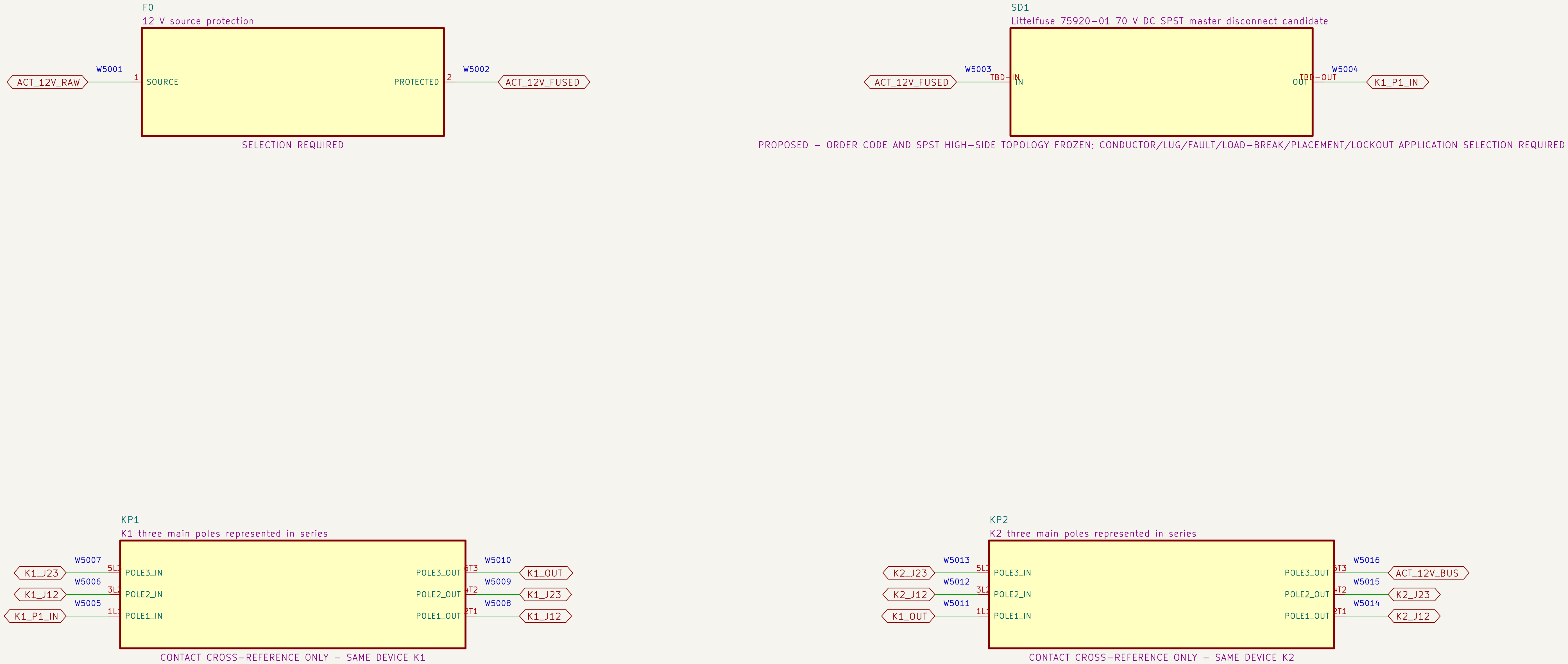
UNACCEPTED SRA1−SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE
Project Button
Sheet: /04 Contactor coils, mirror contacts and EDM/
File: 04_contactor_edm.kicad_sch
Title: PB HR−V0 ELEC P1.21 − 04

Size: A3	Date: 2026−08−11	**Rev: P1.21**
KiCad E.D.A. 10.0.5		Id: 5/13

PRELIMINARY — NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION

05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



NOTE 1: Series jumpers: K1 2T1->3L2, 4T2->5L3, 6T3->K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault-current evidence.

UNACCEPTED SRA1-SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE

Project Button

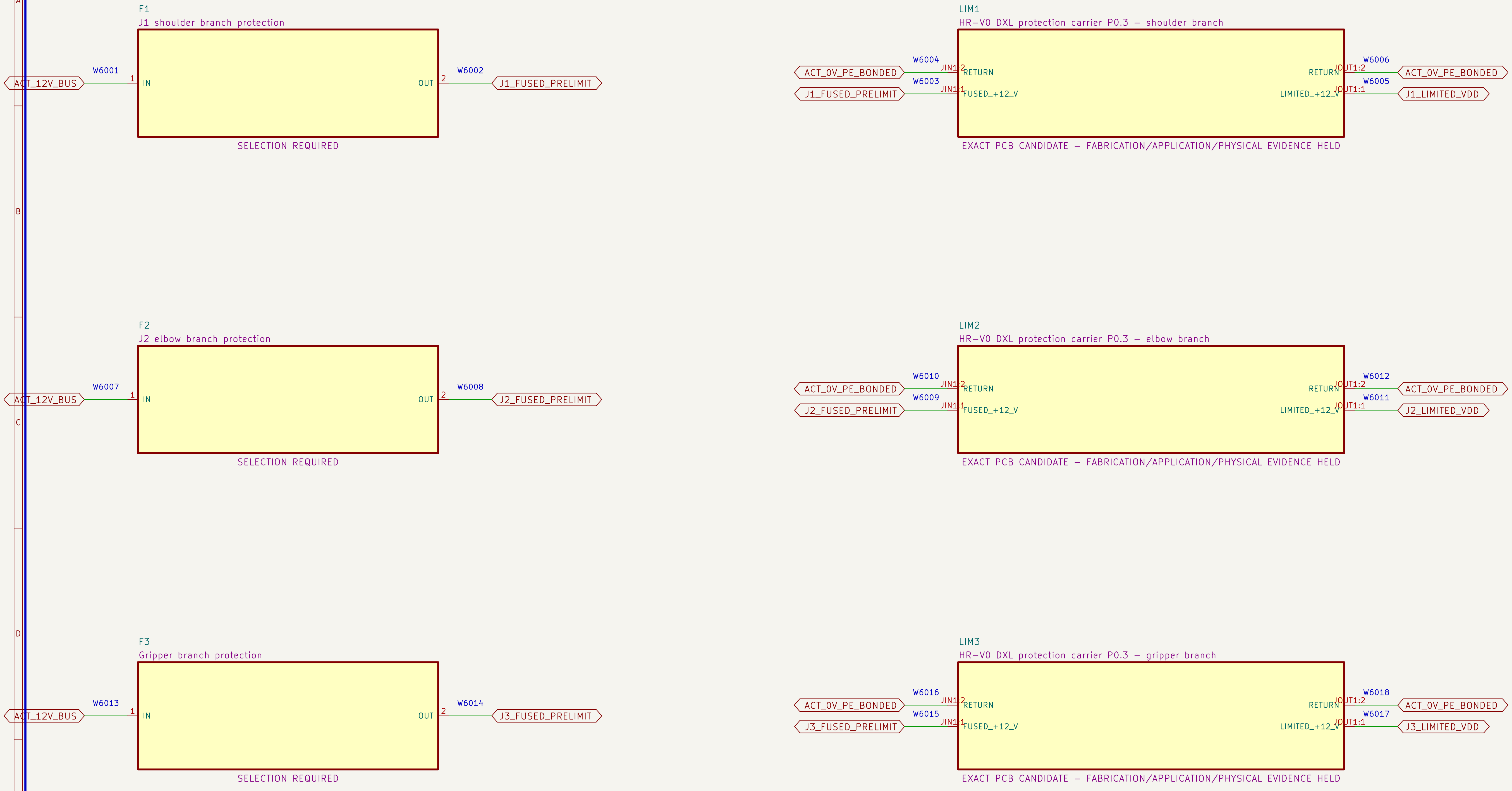
Sheet: /05 Redundant actuator–power interruption/
File: 05_actuator_interruption.kicad_sch

Title: PB HR-V0 ELEC P1.21 — 05

Size: A3	Date: 2026-08-11	**Rev: P1.21**
KiCad E.D.A. 10.0.5		Id: 6/13

06 Protected actuator branches and current–limiter carriers

Each fuse output and carrier output has a distinct positive rail; physical protection performance remains unproved.



NOTE 1: INJ1 is a fixed central star: U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal–integrity, pull and no–backfeed tests remain release gates.

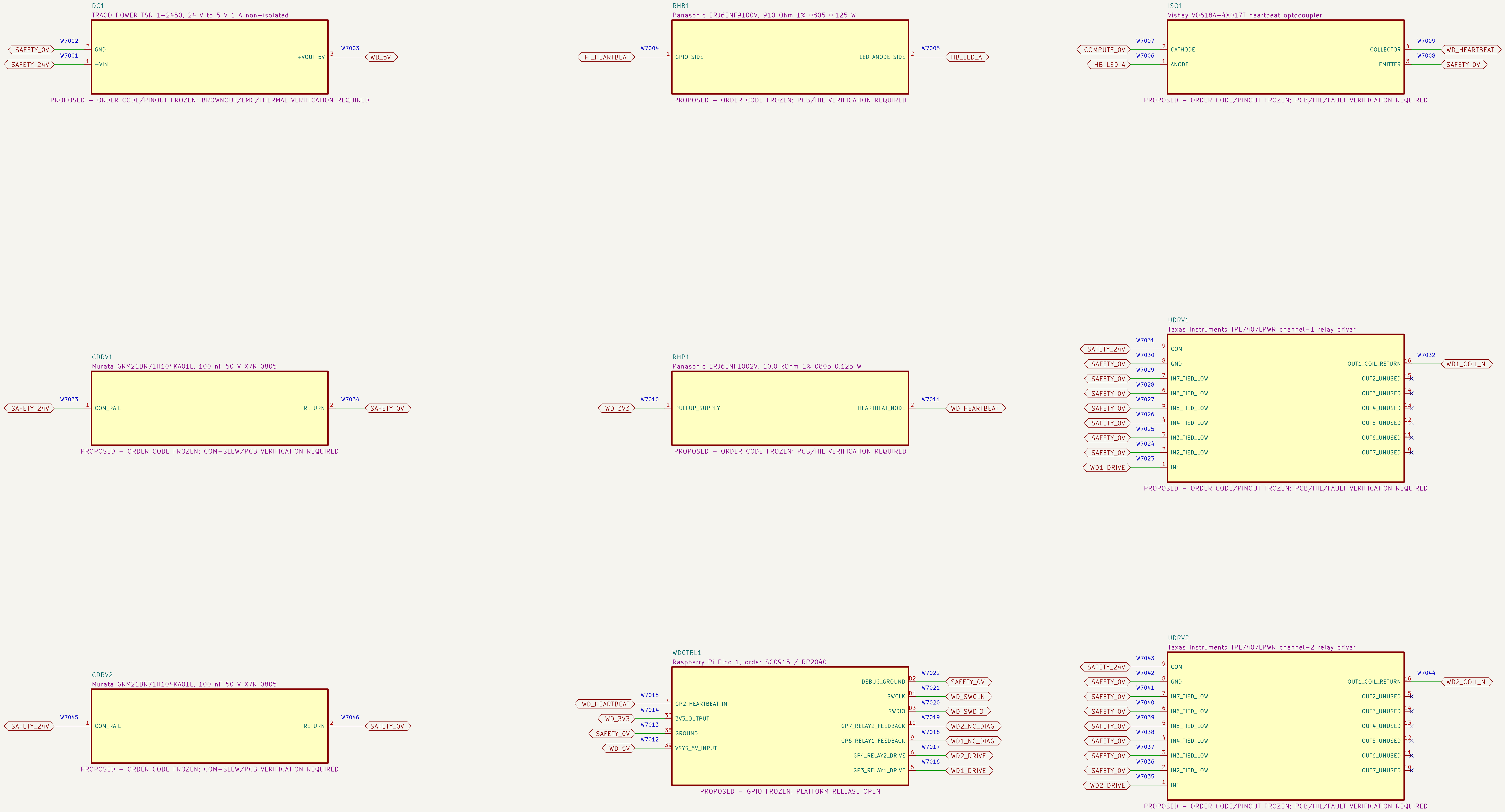
UNACCEPTED SRA1–SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE
Project Button
Sheet: /06 Protected actuator branches and current–limiter carriers/
File: 06_branches_and_limiters.kicad_sch

Title: PB HR–V0 ELEC P1.21 – 06

Size: A3	Date: 2026–08–11	**Rev: P1.21**
KiCad E.D.A. 10.0.5		Id: 7/13

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



NOTE 1: Power–up, brownout, clock, stuck–GPIO and firmware tests are mandatory.

NOTE 2: Qualified review decides whether watchdog loss is credited or diagnostic only;
current safety credit is NONE.

UNACCEPTED SRA1–SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE
Project Button
Sheet: /07 Independent watchdog power, controller and drivers/
File: 07_watchdog_control.kicad_sch

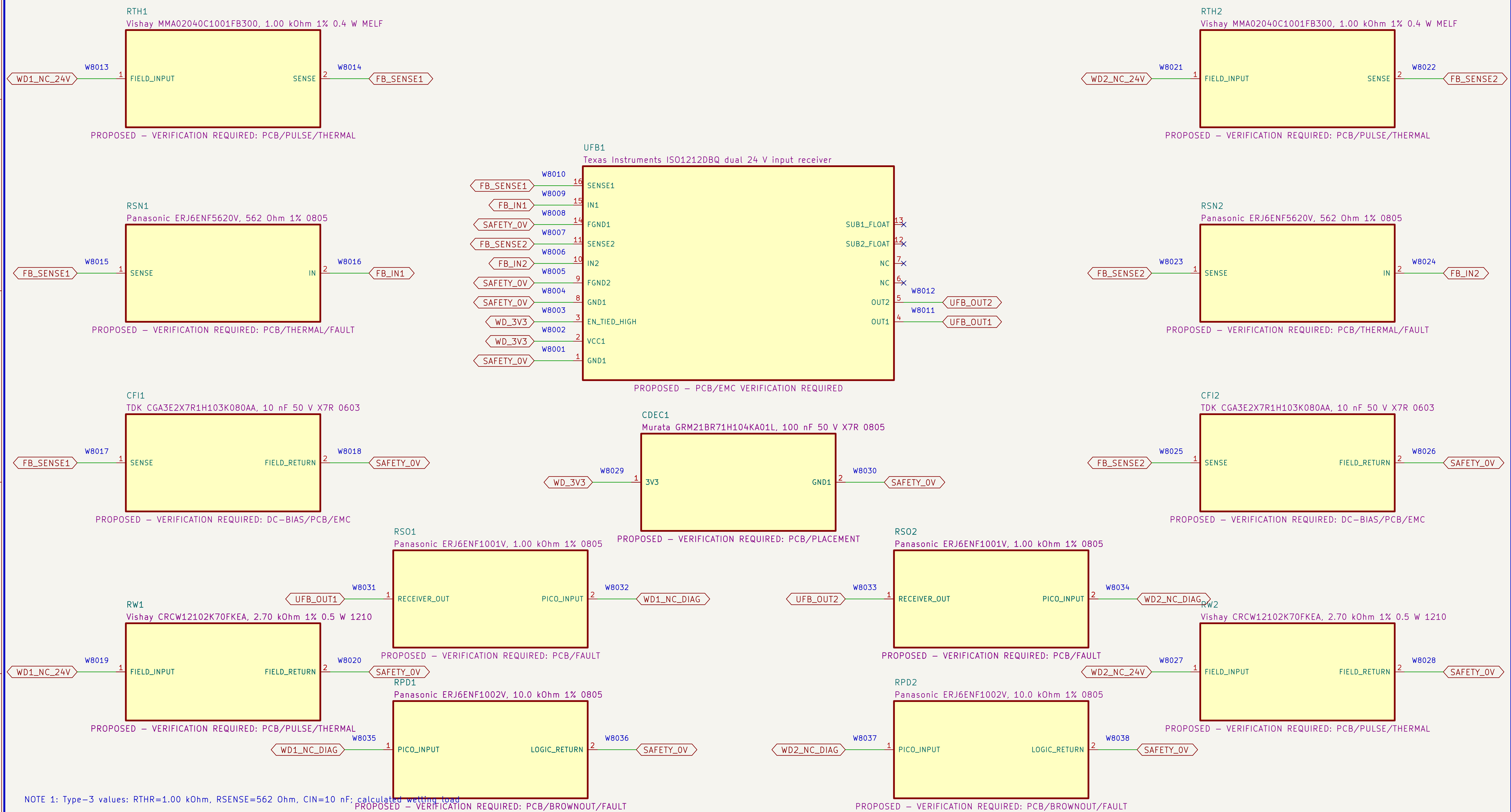
Title: PB HR–V0 ELEC P1.21 – 07

Size: A2 | Date: 2026–08–11 | **Rev: P1.21**
KiCad E.D.A. 10.0.5 | Id: 8/13

PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION

08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



NOTE 1: Type-3 values: RTHR=1.00 kOhm, RSENSE=562 Ohm, CIN=10 nF; calculated wetting load is 2.70 kOhm 1% 0.5 W per channel.

NOTE 2: Both grounds are SAFETY_0V; the ISO1212 barrier is not credited. Passive identities are frozen; PCB, derating, EMC and fault tests remain open.

UNACCEPTED SRA1-SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE

Project Button

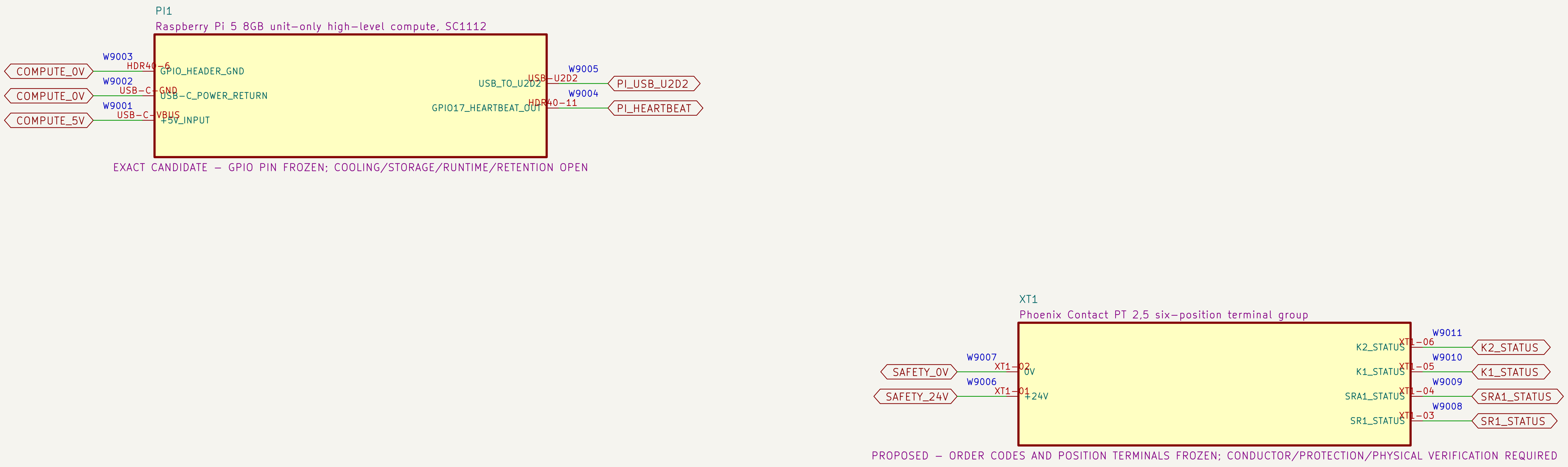
Sheet: /08 Calculated dual-channel 24 V watchdog feedback/
File: 08_watchdog_feedback_interface.kicad_sch

Title: PB HR-V0 ELEC P1.21 – 08

Size: A3	Date: 2026-08-11	**Rev: P1.21**
KiCad E.D.A. 10.0.5		Id: 9/13

09 Compute and control terminals

High–level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2 under a future controlled unpowered–fixture procedure.

NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back–power the board or bypass a protective function.

NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range, torque and enclosure layout remain selection required.

UNACCEPTED SRA1–SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE

Project Button

Sheet: /09 Compute and control terminals/
File: 09_compute_and_control_terminals.kicad_sch

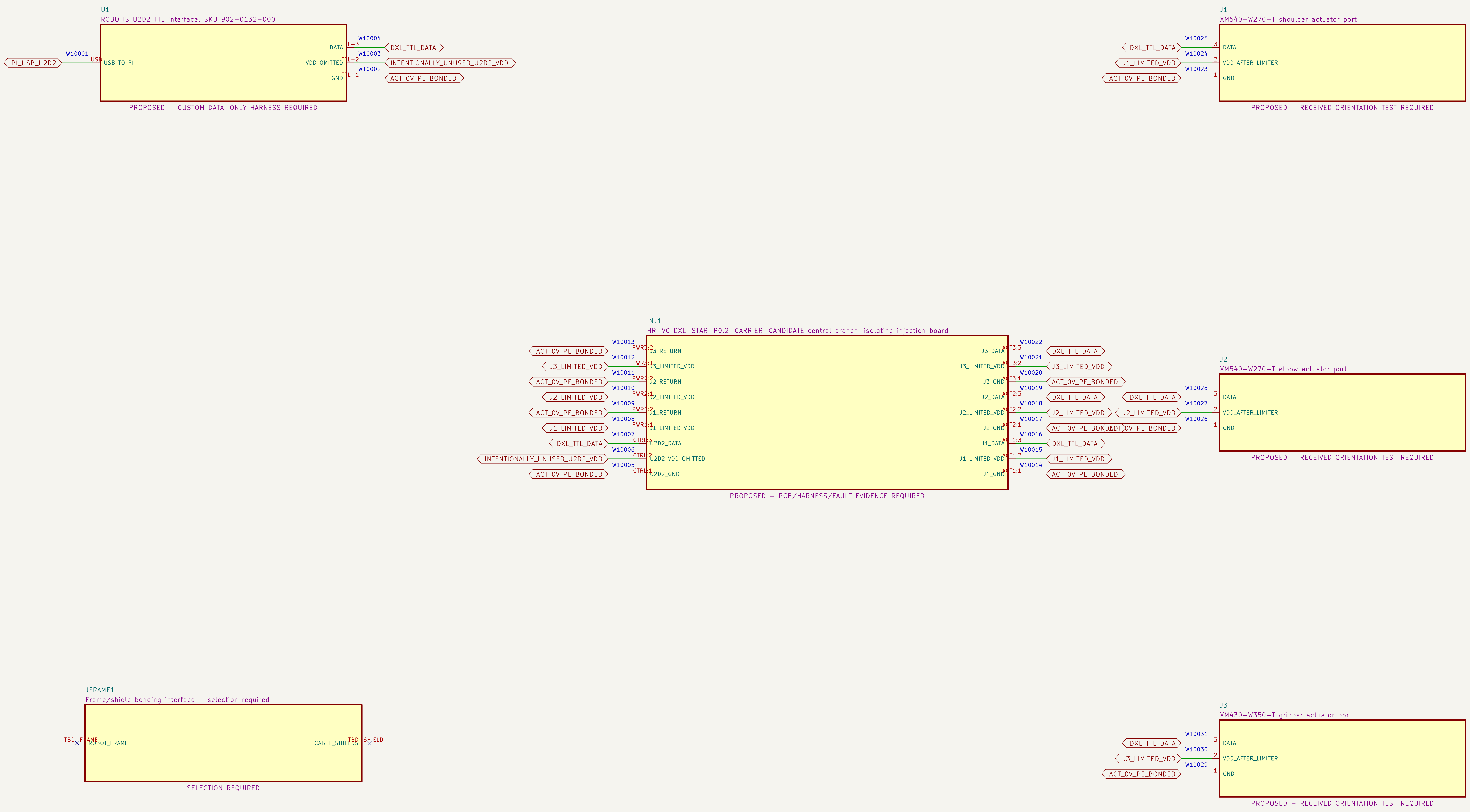
Title: PB HR–V0 ELEC P1.21 — 09

Size: A3	Date: 2026–08–11	**Rev: P1.21**
KiCad E.D.A. 10.0.5		Id: 10/13

PRELIMINARY — NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION

10 U2D2, DXL star, actuator ports and bonding boundary

U2D2 pin 2 is omitted; DXL-STAR accepts three distinct limited positive rails and routes DATA/return.



NOTE 1: The V0 bus is TTL; HR-30 RS-485 remains a separate architecture.

NOTE 2: Do not bond frame or shields to 0V/PE until EMC and parallel-path review accepts the exact implementation.

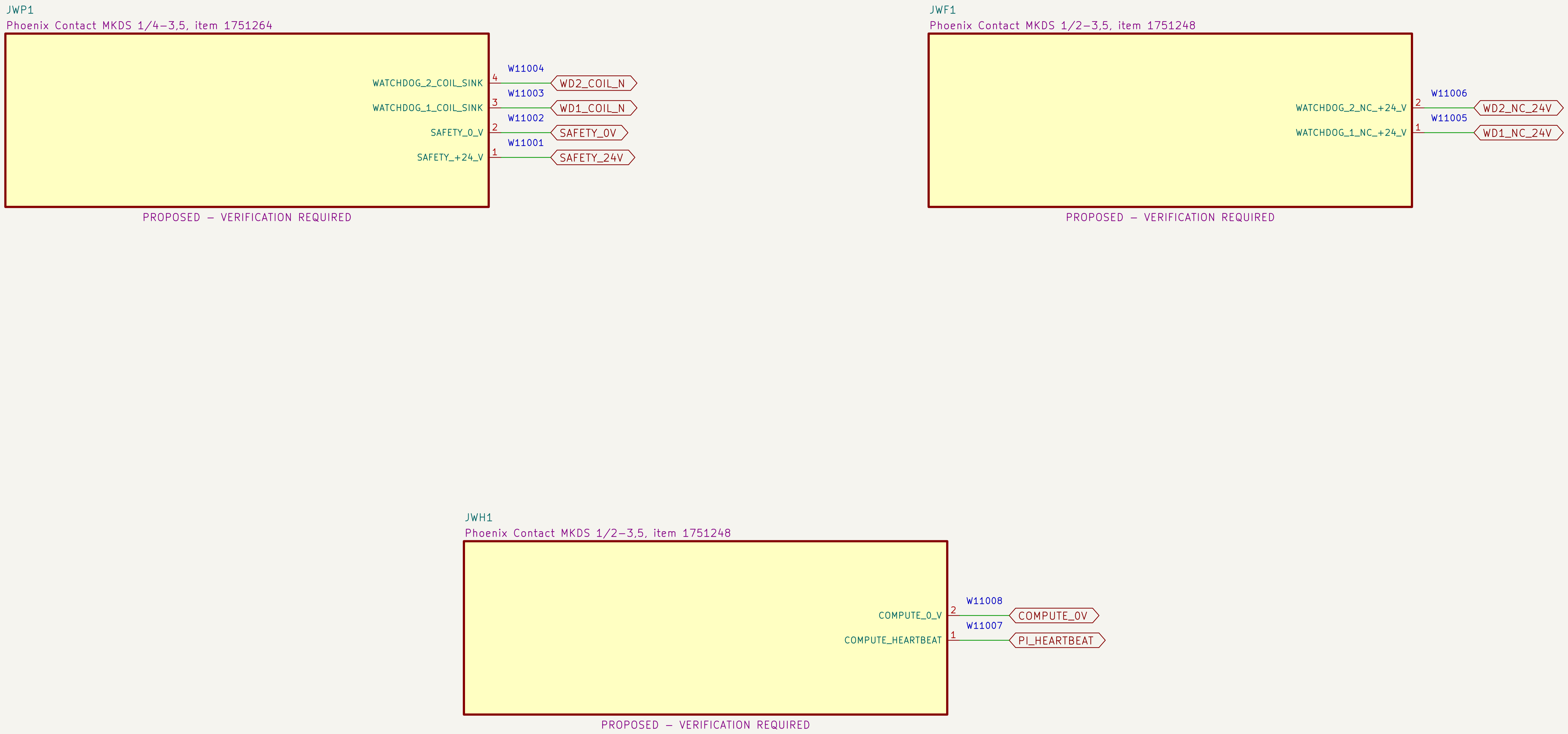
UNACCEPTED SRA1-SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE
Project Button
Sheet: /10 U2D2, DXL star, actuator ports and bonding boundary/
File: 10_actuator_interfaces.kicad_sch

Title: PB HR-V0 ELEC P1.21 - 10

Size: A2 | Date: 2026-08-11 | **Rev: P1.21**
KiCad E.D.A. 10.0.5 | Id: 11/13

11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

UNACCEPTED SRA1–SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE

Project Button

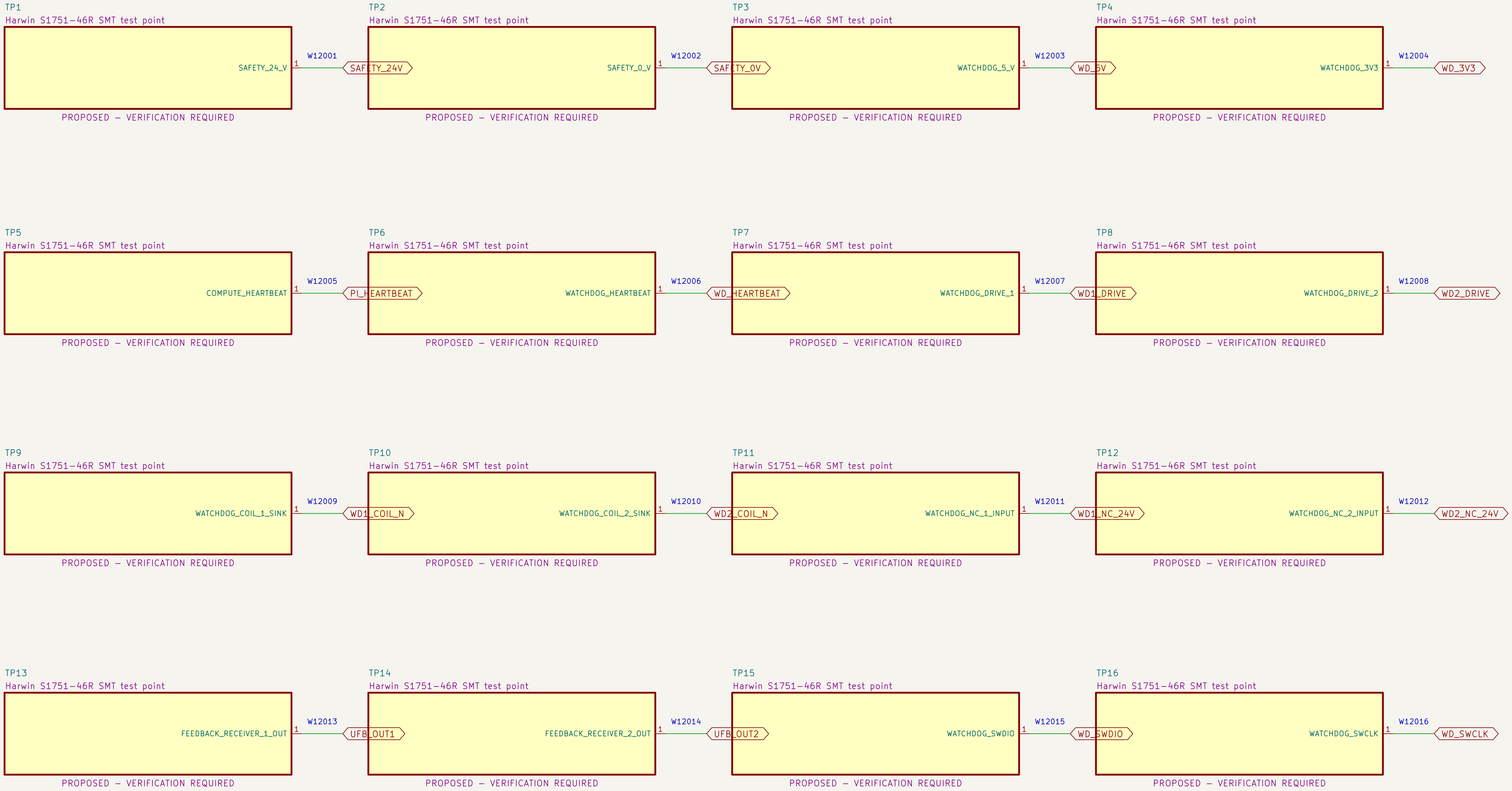
Sheet: /11 Watchdog PCB external connectors/
File: 11_watchdog_pcb_connectors.kicad_sch

Title: PB HR–V0 ELEC P1.21 – 11

Size: A3	Date: 2026–08–11	**Rev: P1.21**
KiCad E.D.A. 10.0.5		Id: 12/13

12 Watchdog PCB test access

Exact clip–compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

UNACCEPTED SRA1–SUPPLY CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE
Project Button
Sheet: /12 Watchdog PCB test access/
File: 12_watchdog_pcb_test_access.kicad_sch
Title: PB HR–V0 ELEC P1.21 – 12
Size: A3 | Date: 2026–08–11 | **Rev: P1.21**
KiCad E.D.A. 10.0.5 | Id: 13/13